Edward Y. Chang
Assistant Professor

Department of Electrical & Computer Engineering
University of California
Santa Barbara, California 93106
Phone: (805) 893-2917
Fax: (805) 893-3262
Email: echang@ece.ucsb.edu
http://www-db.stanford.edu/ echang

EDUCATION

Ph.D.	Electrical Engineering, **Stanford University**	1995-99
M.S.	Computer Science, **Stanford University**	1992-94
M.S.	Industrial Engineering, **University of California, Berkeley**	1984-85
B.S.	Industrial Engineering, **Tunghai University, Taiwan**	1977-81

AWARDS

IBM Faculty Partnership Award 2000
California DiMI Opportunity Award 2000
Digital Equipment Corporation Excellence Award 1989

RESEARCH EXPERIENCE

Assistant Professor, University of California, Santa Barbara 1999–Present

Principle Investigator of the Perception-Based Image Retrieval Project. Supervise four Ph.D. students to investigate image representation, machine learning and data management techniques for personalizing image retrieval. Also conduct research on interactive Digital TV and data broadcasting.

Research Assistant, Stanford University 1995–1999

Advisor: Professor Hector Garcia-Molina.
Member of the Digital Libraries project. Designed a parallel media-server architecture that features fast response time for retrieving media clips while maintaining balanced workload and high throughput. Designed and implemented a memory and disk integrated cache that supports interactive TV operations, such as pause, fast-forward and instant replay, at low cost. Also, co-designed and developed a novel indexing structure for finding similar objects in very high-dimensional spaces efficiently.

Visiting Scientist, NEC Multimedia Laboratories, San Jose, CA 1998-1999

Manager: Dr. Yoshinori Hara.
Visitor of the Hypermedia group. Designed and developed machine learning algorithms for personalizing individuals' bookmark and email organization.

Research Intern, Hewlett-Packard Laboratories, Palo Alto, CA Summer 1998

Mentors: Dr. Meichun Hsu and Dr. Umesh Dayal.

Member of the Data Mining Research Team. Designed and developed a system that mines distributed Web logs to discover access patterns for devising effective Web-page caching strategies.

WORK EXPERIENCE

Staff Engineer, Sun Microsystems, Mountain View, CA 1994–1995

Designed and developed the dynamic typing system for CORBA IDL. (The dynamic type of CORBA IDL, called ANY type, is similar to **XML**.)

Principal Software Engineer, Digital Equipment Corp., Palo Alto, CA 1989-1994

Designed and developed a transaction manager to safeguard the atomicity of distributed transactions among heterogeneous databases. Also represented the corporation in the **XOPEN** transaction-processing standards committee.

Senior Software Engineer, Digital Equipment Corp., Mountain View, CA 1987-1989

Led an engineering team to design and develop an Inter-Process Communication (IPC) facility for the VAX/VMS operating system. The IPC facility pinned and managed a pool of shared physical memory for efficient message passing among the local processes.

Software Engineer, Consilium Inc., Mountain View, CA 1985–1987

Designed and developed software that keeps track of the Work-In-Progress (WIP) in IC fabrication processes, analyzes the collected data, and generates effective inventory control and production plans for improving yields and minimizing cost. Also developed software that controls automation cells consisting of robots, wet benches, furnaces, and other semiconductor fabrication equipments.

PROFESSIONAL ACTIVITIES

Referee ACM SIGMOD, PODS, TOIS ICDE, TKDE, The Computer Journal, and The Computer Communication Journal.
Member ACM and IEEE.

PATENTS

- *"πDTV: A Personalizable and Interactive DTV Architecture,"* E. Chang and R. Rangaswami, University of California, patent pending 2000.
- *"Perception-Based and Personalizable Image Retrieval,"* E. Chang, University of California, patent pending 2000.
- *"Methods for Categorizing and Organizing WWW Document Collections,"* K. Vu, W. Li and E. Chang, NEC Research Labs., patent pending 1999.
- *"Reliable Computing Services: Using logic primitives to model various transaction semantics,"* J. Klein, E. Chang, E. Cheng, et al., Digital Equipment Corporation, patent issued 1992.

WORK SUBMITTED

C-1. *"Data Placement for Interactive Broadcast DTV,"*
R. Rangaswani, E. Chang and C. Li.

C-2. *"Testing Query Containment in the Presence of Limited Access Patterns,"* C. Li and E. Chang.

C-3. *"Classifying Web Documents Considering Personalization,"*
Q. Vu, W. Li and E. Chang.

C-4. *"MEGA - Maximizing Expected Generalization for Learning k-CNF Query Concepts,"* B. Li, E. Chang, H. You, and C. Li.

WORK IN PRESS

B-1. *"Approximate Similarity Queries in High-Dimensional Spaces,"*
C. Li, E. Chang, H. Garcia-Molina, and G. Wiederhold, to appear in IEEE Transaction on Knowledge and Data Engineering, 2001.

B-2. *"On Answering Queries in the Presence of Limited Access Patterns,"*
C. Li and E. Chang, to appear in Proc. of Internation Conference on Database Theory, 2001.

B-3. *"Managing Parallel Disks for Continuous Media Data,"*
E. Chang, C. Li and H. Garcia-Molina, to appear in Information Organization & Databases, pp.107-120, Kluwer Publisher, 2000.

PUBLICATIONS

17. *"Towards Perception-Based Image Retrieval,"*
E. Chang, B. Li and and C. Li, in Proc. of the Workshop on Content-Based Access of Image and Video Libraries, pp.101-105, Hilton Head, South Carolina, June 2000.

16. *"Query Planning with Limited Capability,"*
C. Li and E. Chang, in Proc. of the 16th International Conference on Data Engineering, pp.401-412, San Diego, March 2000.

15. *"Maximizing QoS for Interactive DTV Clients,"*
E. Chang, in The Computer Communication Journal (Special Issue), Volume 23, Number 3, pp.205-218, February 2000.

14. *"πDTV: A Cient-Based Interactive DTV Architecture,"*
E. Chang, in Proc. of the 7th ACM International Conference on Multimedia, Part II, pp.111-114, Orlando, November 1999.

13*. *"Searching Near-Replicas of Images via Clustering,"*
E. Chang, C. Li J. Wang, P. Mork and G. Wiederhold, in Proc. of SPIE Symposium of Voice, Video, and Data Comm., Boston, Sept. 1999.

12*. *"PowerBookmarks: A System for Personalizable Web Information Organization, Sharing, and Management,"*
W. Li, K. Vu, E. Chang, and et al., in Proc. of the ACM SIGMOD Conference, Philadelphia, pp. 565-567, June 1999.

11*. *"MEDIC: A Memory & Disk Cache for Multimedia Clients,"*
E. Chang and H. Garcia-Molina, in Proc. of the 6th IEEE International Conf. on Multimedia Computing and Systems, pp.493-99, Florence, Italy, June 1999.

...PUBLICATIONS

10*. *"On Constructing Personalized Navigation Trees for Web Documents,"*
K. Vu, W. Li and E. Chang, in Proc. of the 8th International WWW Conference, Toronto, May 1999.

9. *"RIME: A Replicated Image Detector for the World-Wide Web,"*
E. Chang, J. Wang, C. Li and G. Wiederhold, in Proc. of SPIE Symposium of Voice, Video, and Data Comm., pp.58-67, Boston, November 1998.

8. *"An Image Coding and Reconstruction Scheme for Mobile Computing,"*
E. Chang, in Proc. of the 5th IDMS, pp.137-48, Oslo, September 1998.

7. "2D BubbleUp - Managing Parallel Disks for Media Servers,"
E. Chang, H. Garcia-Molina and C. Li, in Proc. of the 5th International Conference on Foundations of Data Organization (FODO'98), pp.221-230, Kobe, Japan, November 1998.

6. *"Cost-Based Media Server Design,"*
E. Chang and H. Garcia-Molina, in Proc. of the 8th Research Issues in Data Engineering (RIDE'98), pp.76-83, Orlando, February 1998.

5. *"Reducing Initial Latency in Media Servers,"*
E. Chang and H. Garcia-Molina, in IEEE Multimedia, pp.50-61, vol.4, no.3, 1997.

4. *"BubbleUp: Low Latency Fast-Scan for Media Servers,"*
E. Chang and H. Garcia-Molina, in Proc. of the 5th ACM International Conference on Multimedia, pp.87-98, Seattle, November 1997.

3. *"Effective Memory Use in a Media Server,"*
E. Chang and H. Garcia-Molina, in Proc. of the 23rd Very Large Data Base (VLDB'97) Conference, pp.496-505, Athens, Greece, August 1997.

2. *"Reducing Initial Latency in a Multimedia Storage System,"*
E. Chang and H. Garcia-Molina, in Proc. of the 3rd International Workshop on Multimedia Databases, pp.2-11, Blue Mountain Lake, New York, August 1996.

1. "An Open and Extensible Event-based Transaction Manager,"
E. Cheng, E. Chang, and et al., in Proc. of the Summer USENIX, pp.49-57, Nashville, June 1991.